Microsoft 365 HomeLab – Setup Summary

Objective: Extended the existing lab.test Active Directory environment into Microsoft 365 to practise hybrid identity, cloud administration, endpoint management and IT support.

Tenant and licensing:
Created the Alfie IT Lab tenant using AlfieITLab.onmicrosoft.com. Configured group-based Microsoft 365 Business Premium licensing through M365-Licensed. Confirmed that Alice Morgan and Farah Ahmed could send and receive email. Disabled recurring billing for the trial.

Hybrid identity:
Configured Microsoft Entra Connect Sync on DC01 with organisational-unit filtering, password hash synchronisation and password writeback. Updated synchronised users' sign-in names to use the tenant domain. Verified that selected on-premises accounts appeared in Microsoft 365.

Identity security:
Enabled the LAB - Require MFA Conditional Access policy and verified that Alice needed Microsoft Authenticator to access Outlook. Excluded the emergency administrator account and the Directory Synchronization Accounts role from the lab Conditional Access policies.

Device identity and enrolment:
Verified CLIENT01's hybrid join, with AzureAdJoined and DomainJoined both reporting YES. Configured Group Policy for automatic MDM enrolment, but the successful Intune enrolment was completed manually using the MDM-only enrolment workflow. CLIENT01 appeared in Intune with Alice Morgan as its primary user and Personal ownership.

Intune management:
Assigned LAB - Windows Compliance with a minimum OS version of 10.0.26200.0 and confirmed CLIENT01 was compliant. Deployed an Edge configuration enabling the Home button and verified ShowHomeButton was true with status OK in edge://policy. Deployed Company Portal as a required application and confirmed Installed status in Intune.

Setup difficulties:
CLIENT01 initially appeared in Entra without appearing in Intune. Hybrid device identity and Intune enrolment were checked separately. Manual MDM enrolment succeeded after signing into Windows with the domain administrator account and authenticating the enrolment using Alice's cloud account.

Self-service password reset:

Successfully tested self-service password reset for a synchronised user. The new password was written back to on-premises Active Directory, verified through a fresh Windows domain sign-in and continued access to the Sales share.

Location-based Conditional Access:
Tested the location-based Conditional Access policy using What If. Confirmed that it would block a US sign-in and exclude a UK sign-in. Policy remains in Report-only mode.

Evidence:
HomeLab-Intune-CLIENT01.png
HomeLab-Intune-Compliance.png
HomeLab-Intune-EdgePolicy.png
HomeLab-Intune-CompanyPortal-Installed.png
HomeLab-CompanyPortal-CLIENT01.png

Phase 7 Exercise 1 – Missing Microsoft 365 User

Scenario: Created Jamie Taylor in on-premises Active Directory to simulate a new employee missing from Microsoft 365.

- **Symptom:** Jamie did not appear in Microsoft 365 Active users after a synchronisation cycle.
- **Cause:** His account was created in the default Users container, outside the organisational units selected for Entra Connect synchronisation.
- **Fix:** Moved Jamie's account into LabCompany → Staff → Sales, which was within the synchronisation scope, then triggered a delta synchronisation.
- **Verification:** Jamie appeared in Microsoft 365 Active users as jamie.taylor@AlfieITLab.onmicrosoft.com. His account remained unlicensed; this exercise tested synchronisation only.
- **Evidence:** HomeLab-M365-Admin-JamieTaylorAppeared.png and GLPI ticket #4.
- **Learning:** An AD account must be within Entra Connect's configured synchronisation scope to appear in Microsoft 365. Synchronising an account and assigning a licence are separate steps.

Phase 7 Exercise 2 – MFA recovery after replacing phone

Scenario: Simulated Alice Morgan replacing her phone and losing access to Microsoft Authenticator.

- **Investigation:** Checked Alice's authentication methods in Entra and identified her existing Authenticator registration.
- **Recovery:** Created a short-lived Temporary Access Pass, removed the old Authenticator registration, and used the pass to register Authenticator again. The same physical phone represented the replacement.
- **Cleanup:** Removed the Temporary Access Pass after registration.
- **Verification:** Alice successfully opened Outlook in a fresh private session using her normal password and the newly registered Authenticator.
- **Documentation:** Recorded the resolution in GLPI ticket **#5**; Alice accepted the solution.
- **Learning:** A Temporary Access Pass can restore access for authentication-method registration without disabling the MFA policy.

Phase 7 Exercise 3 – Managing a noncompliant device

Scenario: Simulated a misconfigured Intune policy by raising the minimum Windows version from 10.0.26200.0 to 10.0.30000.0.

- **Symptom:** CLIENT01's Company Portal displayed "Can't access company resources" and requested Windows 10.0.30000.0 or later.
- **Cause:** The policy required a newer OS version than CLIENT01's installed build, 10.0.26200.6584.
- **Fix:** Restored the minimum version to 10.0.26200.0 and ran Check access.
- **Verification:** Company Portal returned to green with "Can access company resources."
- **Evidence:** HomeLab-CLIENT01-Noncompliant.png // HomeLab-CLIENT01-Compliance-Restored.png // HomeLab-GLPI-DeviceCompliance.png // GLPI Ticket #6
- **Learning:** Compliance policies assess device requirements. This exercise verified noncompliance detection and recovery; it did not verify an application being blocked by Conditional Access.

Phase 7 Exercise 4 – Blocked M365 Sign-in and recovery

Scenario: Temporarily enabled **Block sign-in** for Alice Morgan in Microsoft 365.

- **Symptom:** A fresh Outlook sign-in was denied with "Your account has been locked. Contact your support person to unlock it, then try again."
- **Cause:** The deliberately applied cloud account sign-in block.
- **Fix:** Cleared **Block sign-in** and saved the change.
- **Recovery issue:** The first Outlook retry returned a **401 error**. Allowed time for the change and retried in a fresh private browser session.
- **Verification:** Alice successfully signed into Outlook using her normal credentials and Authenticator.
- **Evidence:** HomeLab-Alice-CloudSignIn-Blocked.png // GLPI Ticket #8
- **Learning:** A Microsoft 365 sign-in block is separate from an on-premises AD account lockout. Recovery must be verified through a fresh sign-in.

Phase 7 Exercise 5 – Password sync failure and recovery

Scenario: Simulated a sync outage by stopping the **ADSync** service on DC01, then resetting Alice Morgan's password in local AD.

- **Symptom:** Microsoft 365 rejected the new password but still accepted the previous password.
- **Cause:** The stopped ADSync service prevented the password change from synchronising to Entra ID.
- **Fix:** Ran Start-Service ADSync and confirmed the service returned to **Running**.

- **Verification:** After waiting for password hash synchronisation, Alice successfully opened Outlook in a fresh private session using her new password and MFA.
- **Learning:** Password hash synchronisation runs separately from the normal object-sync schedule. Existing cloud credentials remained usable during this simulated sync outage.
- **Evidence:** GLPI Ticket #9 // HomeLab-PasswordSyncRecovery.png

Phase 7 outcome: All five troubleshooting exercises were reproduced, resolved and verified.